Extracting the ε^0, $\log\delta$-free coefficient of $\delta^{\varepsilon\lambda}$ from a non-diagonalizable residue matrix, without computing a Jordan decomposition

1 Setup

Near a singular point of the differential equation, in the local variable δ (vanishing at that point), the canonical system is Fuchsian:

$$\frac{d}{d\delta}J(\delta) = \frac{\varepsilon}{\delta}A\,J(\delta) \qquad \Longrightarrow \qquad J(\delta) = \delta^{\varepsilon A}J(1) = \exp\bigl(\varepsilon A\log\delta\bigr)\,J(1),$$

where A is a fixed (typically large, here 371×371) rational matrix — the residue of the connection at that singularity. When A is diagonalizable this is trivial: each eigenvector direction just picks up a factor $\delta^{\varepsilon\lambda}$. When A is not diagonalizable (the generic case here), $\exp(\varepsilon A\log\delta)$ also produces terms of the form $\varepsilon^k(\log\delta)^k\delta^{\varepsilon\lambda}$, coming from the nilpotent part of the Jordan blocks. The goal of this note is to isolate, for each eigenvalue $\lambda > 0$, the matrix multiplying $\delta^{\varepsilon\lambda}$ at order ε^0 with *no* accompanying $\log\delta$ — and to do so without constructing A's Jordan basis, which is expensive and numerically fragile at this matrix size.

2 Jordan–Chevalley decomposition and spectral projectors

Every square matrix A over $\mathbb{Q}$ (or $\mathbb{C}$) splits uniquely as

$$A = S + N, \qquad SN = NS, \qquad S \text{ semisimple (diagonalizable)}, \qquad N \text{ nilpotent}.$$

The semisimple part decomposes over the spectrum of A,

$$S = \sum_{\lambda} \lambda\,P_\lambda,$$

where the P_λ are the **spectral projectors**: matrices satisfying

$$P_\lambda^2 = P_\lambda, \qquad P_\lambda P_\mu = 0\ (\lambda\neq\mu), \qquad \sum_\lambda P_\lambda = I, \qquad [P_\lambda, A] = 0,$$

with P_λ projecting onto the generalized eigenspace $\ker(A-\lambda I)^{a_\lambda}$ (a_λ = algebraic multiplicity of λ). Write $N_\lambda := (A-\lambda I)P_\lambda$; this is nilpotent on the image of P_λ, with nilpotency index $m_\lambda \le a_\lambda$ equal to the size of the largest Jordan block at λ.

3 Why P_λ answers the question

Since S and $N = \sum_\lambda N_\lambda$ commute,

$$\exp(\varepsilon A \log\delta) = \exp(\varepsilon S \log\delta)\exp(\varepsilon N \log\delta) = \sum_\lambda \delta^{\varepsilon\lambda}\exp(\varepsilon N_\lambda \log\delta)\, P_\lambda,$$

using $\exp(\varepsilon S\log\delta) = \sum_\lambda \delta^{\varepsilon\lambda} P_\lambda$ and the orthogonality $P_\lambda P_\mu = 0$ to kill all cross terms. Expanding the finite (nilpotent) series

$$\exp(\varepsilon N_\lambda \log\delta) = I + \varepsilon N_\lambda \log\delta + \tfrac{1}{2}(\varepsilon N_\lambda \log\delta)^2 + \dots,$$

the term multiplying $\delta^{\varepsilon\lambda}$ with no $\log\delta$ and at order ε^0 is precisely

$$\boxed{P_\lambda.}$$

So the entire problem reduces to computing, for each positive integer eigenvalue λ of A, the spectral projector P_λ — and doing so without ever building the similarity transform that diagonalizes/Jordanizes A.

4 P_λ as a polynomial in A

Let $\mu_A(z) = \prod_k (z-\lambda_k)^{m_k}$ be the minimal polynomial of A (m_k = Jordan index at λ_k, as above). By the Chinese Remainder Theorem applied to $\mathbb{Q}[z]/(\mu_A(z)) \cong \bigoplus_k \mathbb{Q}[z]/(z-\lambda_k)^{m_k}$, there is a unique polynomial $p_\lambda(z)$, modulo $\mu_A(z)$, such that

$$p_\lambda(z) \equiv 1 \pmod{(z-\lambda)^{m_\lambda}}, \qquad p_\lambda(z) \equiv 0 \pmod{(z-\lambda_k)^{m_k}} \text{ for all } k \neq \lambda,$$

and then

$$P_\lambda = p_\lambda(A).$$

Fact. This is standard matrix functional calculus: $(A-\lambda_k I)^{m_k}$ annihilates the entire generalized eigenspace of λ_k, so $p_\lambda(A)$ vanishes there whenever $k \neq \lambda$ (since p_λ carries that factor); and $p_\lambda(A) - I$ is a polynomial multiple of $(A-\lambda I)^{m_\lambda}$, which annihilates the nilpotent operator on λ's generalized eigenspace, so $p_\lambda(A) = I$ there.

A practical simplification. Computing the exact Jordan index m_k requires extra rank computations of $(A-\lambda_k I)^j$ for increasing j. This can be skipped entirely: using the *algebraic multiplicity* $a_k \geq m_k$ in place of m_k everywhere is always safe, since $(z-\lambda_k)^{a_k}$ is still divisible by $(z-\lambda_k)^{m_k}$ — it only makes p_λ a higher-degree polynomial than strictly necessary. The a_k's are cheaply available: compute the eigenvalues of A numerically, round to the nearest integer, tally multiplicities, and confirm exactness via $\mathrm{Tr}(A), \mathrm{Tr}(A^2), \mathrm{Tr}(A^3), \dots$ matching the power sums predicted by the candidate integer spectrum (these traces are computable exactly and cheaply even for large A).

5 Constructing $p_\lambda(z)$: the role of $q(z)$

Split $p_\lambda(z) = q_\lambda(z)\, g_\lambda(z)$ with

$$g_\lambda(z) := \prod_{k\neq\lambda} (z-\lambda_k)^{a_k}.$$

This single polynomial already satisfies all the "$\equiv 0$" conditions for every $k \neq \lambda$ at once, being their literal product. One condition remains: we need

$$q_\lambda(z)\, g_\lambda(z) \equiv 1 \pmod{(z-\lambda)^{a_\lambda}} \quad \Longleftrightarrow \quad q_\lambda(z) \equiv \frac{1}{g_\lambda(z)} \pmod{(z-\lambda)^{a_\lambda}}.$$

One route to q_λ is the extended Euclidean algorithm on $g_\lambda(z)$ and $(z-\lambda)^{a_\lambda}$, producing a Bézout identity $u(z)g_\lambda(z) + v(z)(z-\lambda)^{a_\lambda} = 1$ and taking $q_\lambda = u$. But because one of the two moduli here is a pure power of a linear factor, and $g_\lambda(\lambda) \neq 0$ by construction (λ is excluded from the product defining g_λ), the function $1/g_\lambda(z)$ is analytic (no pole) at $z = \lambda$ and has an ordinary Taylor series there. So q_λ can be read off directly:

$$q_\lambda(z) = \text{Taylor series of } \frac{1}{g_\lambda(z)} \text{ about } z = \lambda, \text{ truncated at order } a_\lambda - 1.$$

Concretely, substituting $w = z - \lambda$:

$$q_\lambda(z) = \sum_{j=0}^{a_\lambda - 1} c_j\, w^j, \qquad \frac{1}{g_\lambda(\lambda + w)} = \sum_{j=0}^{\infty} c_j\, w^j + O(w^{a_\lambda}).$$

This is the *same* object the Euclidean algorithm would produce — the Euclidean recursion against a pure prime power collapses exactly to a power-series inversion — but Taylor expansion gets there in one step instead of running the recursion.

6 Evaluating on the matrix

- $g_\lambda(A) = \prod_{k \neq \lambda} (A - \lambda_k I)^{a_k}$: each factor computed via fast binary-exponentiation matrix power ($O(\log a_k)$ matrix multiplications), then the factors multiplied together.
- $q_\lambda(A) = \sum_{j=0}^{a_\lambda - 1} c_j (A - \lambda I)^j$: a polynomial of degree $< a_\lambda$ in $B := A - \lambda I$, evaluated by Horner's rule (sequential multiplication by B).
- $P_\lambda = q_\lambda(A) \cdot g_\lambda(A)$.

Every step is a matrix product or matrix power of A (shifted by scalars) — no eigenvectors, no change-of-basis matrix, no matrix inverse anywhere. This is what makes the computation fast and exact even for a 371×371 defective matrix, where building an actual Jordan basis would be far more expensive and numerically delicate.

7 Verification

All computations were carried out in exact rational arithmetic. For every eigenvalue treated:

$$\operatorname{Tr}(P_\lambda) = a_\lambda \text{ (exactly)}, \qquad P_\lambda^2 - P_\lambda = 0 \text{ (the exact zero matrix)}.$$

Obtaining an exact zero residual (not merely small) across a 371×371 matrix of rational numbers is strong evidence of correctness: an incorrect construction would not cancel to exactly zero by chance.

8 Results

The method was applied to four residue matrices, each 371×371, with integer spectra confirmed exactly via trace matching.

File	Spectrum (eigenvalue : multiplicity)	Positive eigenvalues processed
`toexp1.m`	-4:2, -3:132, -2:29, -1:26, 0:77, 1:96, 2:5, 3:4	$1, 2, 3$
`toexp2.m`	identical spectrum to `toexp1.m`	$1, 2, 3$
`toexp3.m`	-7:4, -6:41, -5:10, -4:21, -3:44, -2:73, -1:33, 0:118, 1:2, 2:23, 3:2	$1, 2, 3$
`toexp7.m`	-7:2, -4:4, -3:66, -2:1, 0:128, 1:153, 2:5, 3:10, 5:2	$1, 2, 3, 5$

All thirteen resulting projector matrices P_λ (one file per (matrix, eigenvalue) pair, named `P_<lambda>_toexp<k>.m`) passed the trace and idempotency checks above and were exported to `matrix_to_JD/toJD/`.